Mahika Gunjkar

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EDUCATION

Carnegie Mellon University, Pittsburgh, PA **Dec 2026**
Master of Information Systems Management – Business Intelligence & Data Analytics
Coursework: Intro to Deep Learning, Unstructured Data Analytics, Machine Learning for Problem Solving, Advanced Business Analytics, Statistics

University of Cincinnati, Cincinnati, OH **May 2025**
Bachelor of Business Administration – Information Systems | Certificate: Business Analytics
Coursework: Data Mining, Business Intelligence, Forecast Risk & Analysis, Advanced App Development, IT Architecture & Networks

TECHNICAL SKILLS

- **Data Analysis & Coding:** Python, R, SQL/PostgreSQL, Oracle DBMS, Pandas, Tableau, Power BI, Jupyter Notebook, Weights & Biases, React.js, TypeScript, JavaScript, Tailwind CSS, Chart.js, Tableau, Power BI, KPI Tracking, EDA
- **Machine Learning:** Scikit-learn, Regression & Clustering Models, Feature Engineering, Hyperparameter Tuning, Forecasting & Risk Modeling, Statistical Analysis
- **Data Engineering & Cloud:** AWS (EC2, G4), Docker, Git, ETL Pipelines (Python), Flask, MongoDB, NumPy

WORK EXPERIENCE

Research Assistant – CMU PLUS Lab | Carnegie Mellon University **May 2026 – Present**
- Developing a React and Python web application enabling middle school math teachers to monitor student progress and set weekly learning goals from i-Ready assessment data, delivering actionable performance insights at scale
- Engineering ETL pipelines to process and integrate multi-source student learning datasets into teacher-facing dashboards with automated weekly progress reports, reducing manual reporting overhead significantly
- Supporting AWS cloud infrastructure deployment including authentication workflows, email automation, and production server configuration to ensure robust, university-grade system availability

Software Development Intern | Revdau Industries **May 2024 – July 2024**
- Engineered regression and clustering models on internal KPIs through systematic feature engineering, cross-validation strategy, and iterative hyperparameter optimization
- Architected and managed scalable data storage solutions on AWS (EC2, G4), maintaining 99% uptime and enabling seamless, low-latency data accessibility for cross-functional development and analytics teams
- Developed automated ETL pipelines in Python reducing manual data preparation time by 25% via optimized scripting, vectorized transformations, and reusable advanced data structures

Resource Management Group Intern | Tata Consultancy Services **May 2023 – July 2023**
- Allocated and aligned 100+ employees to high-priority engagements by analyzing technical skill matrices against organizational demand-capacity models, optimizing resource utilization rates and minimizing bench time
- Designed and automated daily and weekly workforce tracking reports, improving senior leadership visibility and reducing reporting latency, enabling faster, data-driven resourcing decisions

PROJECT EXPERIENCE

EasyTrip – Predictive Travel Planner | Full Stack Web Application **Fall 2025**
- Developed a 12-month predictive flight price visualizer using Chart.js, empowering travelers with data-driven booking decisions grounded in historical price trend analysis and seasonal demand modeling
- Built a Flask-based single-page application integrating 3+ external APIs, reducing average trip planning time from 3 hours to 15 minutes through intelligent workflow automation
- Implemented heuristic recommendation algorithms to surface hotel and transit options dynamically based on user-defined budget constraints and preference parameters, delivering a personalized planning experience

Scottyscope – ESG Auditing Platform with GenAI | TartanHacks 2026 **February 2026**
- Architected an AI-powered ESG auditing pipeline using Gemini 1.5 Flash that decouples quantitative risk scoring from qualitative insights, eliminating LLM hallucinations and ensuring mathematically verified sustainability reporting
- Integrated the GenAI backend with a responsive React.js frontend to deliver real-time ESG feedback loops, interactive data visualizations, and actionable technical roadmaps for corporate sustainability officers

Latent Denoising Diffusion Probabilistic Model | PyTorch **Feb 2026 – Present**
- Built a Latent Diffusion Model (LDM) from scratch in PyTorch using a VAE for latent-space compression of ImageNet-100; implemented DDPM and DDIM schedulers for high-quality image synthesis and rapid inference